Equity Valuation: DCF and Comparables

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1 The Core Idea

A stock price is a claim on future cash flows. A valuation model turns assumptions about those cash flows and the discount rate into a value per share, which can be compared with the market price. The toolkit does this for 10 large US stocks with two standard methods and checks how much the answer depends on its inputs. For a risk manager the useful output is less the value itself than how fragile it is.

- **DCF:** project free cash flow, discount it at the WACC, add a terminal value.
- **Comparables:** apply the median EV/EBITDA multiple of the group to each company's EBITDA.

The results show the limits of the methods as much as they show mispricing. A gap of -90% points to a mismatch between the model and the company, not to a trading signal.

2 Why It Matters

- **The terminal value carries the answer.** In the worked example it is 75.0% of enterprise value. The Gordon growth formula (Gordon, 1959) multiplies the last cash flow by $(1+g)/(\text{WACC}-g)$, which is 44.4 for XOM (WACC 4.8%) and 8.2 for NVDA (WACC 15.0%). The same model treats the two firms very differently.
- **Discount-rate errors are not symmetric.** For NVDA, one point less of WACC raises the value by 8.3% and one point more lowers it by 7.1%. A low WACC is more dangerous than a high one, because the value rises faster than it falls.
- **A gap to the market tests the model and the market together.** The DCF is below the price for 5 of 9 stocks, with a median gap of -52.6%. A market that is wrong on five large caps is a less natural reading than a model that cannot express their expected growth.
- **Comparables are relative, not absolute.** They say whether a stock is cheap against its group, not whether the group is cheap. MSFT's comparables gap is zero by construction, because its own multiple is the median.

3 Assumptions

Every item below is built into `equity_valuation.py`. The last sentence of each says what moves if it is wrong.

- **Data.** A one-time Yahoo Finance download (price, shares, market cap, beta, free cash flow, revenue growth, debt, cash, EBITDA, enterprise value). The repository does not record the download date, and the prices are those of that day. A missing beta is replaced by 1.0, a missing revenue growth by 5%, and missing debt or cash by zero. A default beta of 1.0 gives a cost of equity of 9%.
- **Common rates.** One risk-free rate (4%), one market risk premium (5%), one cost of debt (5%) and one tax rate (21%) for every company. A premium of 6% instead of 5% would raise every cost of equity by β points and cut the values, the most for the highest-beta stocks.
- **Capital structure.** Weights use the market capitalisation and Yahoo's total debt. Any debt-like item that figure leaves out is missing from the weights, which makes the debt weight too low and the WACC too high.
- **Starting cash flow.** The last reported free cash flow is taken as given. Yahoo's figure is operating cash flow less capital expenditure, which is already net of interest, yet the code discounts it at the WACC as if it were unlevered. For a firm with a lot of debt this understates enterprise value. A firm with a negative cash flow gets no value.
- **Growth.** The free cash flow grows at the reported revenue growth, clipped to $[-5\%, 20\%]$, fading linearly to 2.5% in year 5. This assumes constant margins and capital intensity. A firm growing faster than 20% is understated by construction.
- **Terminal value.** A Gordon perpetuity at 2.5% growth after year 5, with cash flows at year end. Year-end discounting is conservative: a mid-year convention would raise the value by about $(1 + \text{WACC})^{1/2} - 1$, 4.3% in the worked example.
- **Equity bridge.** Equity value is enterprise value less total debt plus total cash, divided by the basic share count. All cash is treated as excess, and there are no minorities, options or dilution.
- **Comparables.** One median EV/EBITDA over all stocks with data (nine of ten, as JPM has no EBITDA), across sectors, using the trailing EBITDA. The stock being valued is inside its own median. Mixing energy, tech, staples and health care puts a technology-weighted multiple on every firm.
- **Sensitivity.** Only the largest stock by market cap (NVDA), on a 5×5 grid of WACC (base ± 2 points) and terminal growth (2.5% ± 1 point). It shows the effect of two inputs, not of the growth path.

4 Method

4.1 Cost of Capital

The code (`equity_valuation.py`) uses a risk-free rate of 4%, a market risk premium of 5%, a flat cost of debt of 5% and a tax rate of 21%. With the Yahoo Finance beta β, the cost of equity follows the capital asset pricing model (CAPM, Sharpe, 1964):

$$k_e = r_f + \beta \cdot \text{MRP}, \qquad \text{WACC} = \frac{E}{E+D}\, k_e + \frac{D}{E+D}\, k_d(1-t). \tag{1}$$

E is the market capitalisation and D the total debt. The WACC rises with beta and with the equity weight, since equity costs more than after-tax debt (3.95%). A beta of 2.2 or more is needed to reach NVDA's WACC of 15.0%, because $k_e = 4\% + 5\% \cdot \beta$ and the WACC cannot exceed k_e.

4.2 DCF

The starting free cash flow F_0 comes from Yahoo Finance. The growth rate fades linearly over 5 years from the reported revenue growth g_1 (clipped to $[-5\%, 20\%]$) to the terminal growth $g = 2.5\%$:

$$g_t = g_1 + \frac{t-1}{4}(g - g_1), \qquad F_t = F_{t-1}(1 + g_t), \quad t = 1, \ldots, 5. \tag{2}$$

Cash flows are discounted at year end. The Gordon terminal value, the enterprise value and the value per share are

$$TV = \frac{F_5(1+g)}{\text{WACC} - g}, \qquad EV = \sum_{t=1}^{5} \frac{F_t}{(1+\text{WACC})^t} + \frac{TV}{(1+\text{WACC})^5}, \tag{3}$$

$$\text{Value per share} = \frac{EV - (\text{debt} - \text{cash})}{\text{shares outstanding}}. \tag{4}$$

The model returns no value when $F_0 \leq 0$ or missing. Two things push the number up: a higher g_1, which compounds into F_5 and then into TV, and a smaller WACC$-g$. The WACC enters twice, in the discount factors and in the denominator of TV.

4.3 Comparables

The implied price of each stock is

$$P^{\text{comps}} = \frac{m^* \cdot \text{EBITDA} - \text{net debt}}{\text{shares}}, \tag{5}$$

where m^* is the median of EV/EBITDA over the group. The number moves one for one with the multiple and with EBITDA, and the net debt takes the financing back out. No growth, margin or risk difference between firms enters.

4.4 Sensitivity

For the largest stock by market capitalisation (NVDA), the DCF is recomputed on a grid of 5 WACC values (base ± 2 points) and 5 terminal growth values (2.5% ± 1 point). Cells with WACC $\leq g$ are left empty. The grid is a one-at-a-time test of the two inputs the model is most exposed to.

5 Method Comparison

Method	Strength	Weakness
DCF	Explicit about growth, risk and horizon	Terminal value dominates; WACC$-g$ can be tiny
Comparables	Market-based, no discount rate	Needs similar firms; the group can be wrong
Sensitivity grid	Shows how fragile a value is	Varies two inputs only, one pair at a time

Use the DCF when the cash flows are stable enough to forecast and the question is what the business is worth. Use comparables when the question is relative: which of several similar firms is cheap. When the two disagree sharply, as for GOOGL, treat the disagreement as information about the inputs before treating it as a view on the stock. The grid should accompany any DCF number.

6 Results

Valuation screen from `output/valuation_screen.csv`. Prices are in dollars as downloaded.

Ticker	Price	WACC	DCF value	DCF upside	Comps value	Comps upside
XOM	153.04	4.8%	320.69	+109.5%	310.01	+102.6%
UNH	407.08	6.6%	545.97	+34.1%	471.69	+15.9%
PG	145.79	5.7%	167.40	+14.8%	195.25	+33.9%
JNJ	259.24	5.1%	296.25	+14.3%	268.73	+3.7%
AAPL	313.33	9.3%	148.66	−52.6%	221.55	−29.3%
KO	87.05	5.5%	38.87	−55.3%	70.09	−19.5%
GOOGL	354.30	10.0%	97.54	−72.5%	592.78	+67.3%
NVDA	223.96	15.0%	24.12	−89.2%	134.11	−40.1%
MSFT	499.99	9.3%	39.73	−92.1%	499.99	0.0%

- **Growth stocks:** NVDA, MSFT, GOOGL and AAPL come out far below the market price. A 5-year fade to 2.5% growth cannot reproduce the growth the market expects.
- **Low WACC:** XOM has a WACC of 4.8%, only 2.3 points above the terminal growth rate. This inflates the terminal value and the upside.
- **MSFT in comps:** its upside is zero by construction, since its own multiple is the group median.
- **JPM:** no value. Yahoo Finance gives no usable free cash flow or EBITDA for a bank, and the methods do not suit banks.

The DCF gap is bimodal. Four stocks have positive upside (XOM, UNH, PG, JNJ) and five have negative upside, with a mean of −21.0% and a median of −52.6%. The split follows the WACC: the four stocks with positive upside have WACCs from 4.8% to 6.6%, and the four growth names with the largest losses have WACCs from 9.3% to 15.0%. The WACC depends mostly on beta, so part of the ranking is a ranking of betas. KO is the exception: its WACC is 5.5%, yet its DCF is −55.3%, so a low discount rate cannot rescue a cash flow that is too small for the price.

The two methods agree on the sign for 7 of 9 stocks. MSFT is counted as a disagreement: its comparables upside is zero by construction (a rounding residue of 3×10^{-6}%), against −92.1% on the DCF, and GOOGL is the other one. Leaving MSFT out, the count is 7 of 8. The Spearman rank correlation between the two gaps is 0.58 (recomputed from the CSV, all nine stocks). Agreement is weaker evidence than it looks. XOM is up about 100% on both, but for different reasons: the DCF because of a 44-times terminal multiple, the comparables because the group median multiple is far above XOM's own. GOOGL is the opposite case, with −72.5% on the DCF and +67.3% on comparables, a gap of 140 points that the code cannot attribute to either input.

NVDA sensitivity (`output/sensitivity_NVDA.csv`), value per share in dollars:

WACC \ g	1.5%	2.0%	2.5%	3.0%	3.5%
13.0%	26.46	27.44	28.53	29.71	31.03
14.0%	24.42	25.24	26.13	27.11	28.17
15.0%	22.68	23.37	24.12	24.93	25.82
16.0%	21.18	21.77	22.41	23.09	23.83
17.0%	19.87	20.39	20.93	21.52	22.15

The value runs from \$19.9 to \$31.0 over the whole grid, and the most favourable cell is 13.9% of the price of \$223.96. No pair of WACC and terminal growth in the grid closes that gap, so it sits in the cash-flow inputs (the 20% growth cap, the five-year fade, the starting free cash flow) and not in the discount rate. The row labels are rounded: the grid is centred on NVDA's computed WACC of 15.05%.

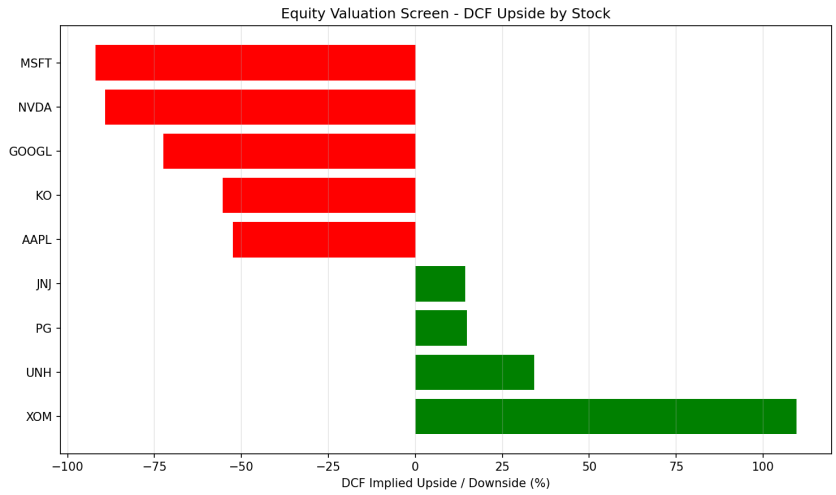


Figure 1: DCF implied upside or downside by stock.

7 Worked Example

The inputs are illustrative: $F_0 = 100$, $g_1 = 10\%$, beta 1.2, market cap 800, debt 200, cash 50, 10 shares.

WACC. $k_e = 4\% + 1.2 \cdot 5\% = 10\%$. Weights are $800/1000 = 0.8$ and 0.2. WACC $= 0.8 \cdot 10\% + 0.2 \cdot 5\% \cdot (1 - 0.21) = 8.79\%$.

Cash flows. The growth path is 10%, 8.125%, 6.25%, 4.375%, 2.5%, which gives $F_{1..5} =$ 110.00, 118.94, 126.37, 131.90, 135.20.

Present value. The discount factors 1.0879^{-t} are 0.9192, 0.8449, 0.7767, 0.7139, 0.6562, so $\sum F_t\, DF_t = 482.64$.

Terminal value. $TV = 135.20 \cdot 1.025/(0.0879 - 0.025) = 2203.14$. Its present value is $2203.14 \cdot 0.6562 = 1445.76$.

Equity value. $EV = 482.64 + 1445.76 = 1928.40$. Net debt is $200 - 50 = 150$. Value per share $= (1928.40 - 150)/10 = 177.84$.

The terminal value is 75.0% of EV. Each input moves the value per share as follows, the others held at their base value:

WACC (%)	6.79	7.79	8.79	9.79	10.79
Value	268.57	214.62	177.84	151.15	130.91
g_1 (%)	0	5	10	15	20
Value	138.77	157.53	177.84	199.79	223.44
g (%)	1.5	2.0	2.5	3.0	3.5
Value	153.42	164.73	177.84	193.21	211.48
Beta	0.8	1.0	1.2	1.5	
Value	244.23	206.19	177.84	146.67	

The terminal value share is 82% at a WACC of 6.79% and 69% at 10.79%, so the lower the discount rate, the more the answer is the terminal value. A starting growth of 35% would be clipped to 20% and give the same 223.44. Matching the comps value of 225 (below) would need a WACC of 7.56%.

Comps. With an illustrative median multiple of 12× and EBITDA of 200, the implied price is $(12 \cdot 200 - 150)/10 = 225$. Each turn of the multiple moves the price by 20, so 8× gives 145 and 16× gives 305. In this example the two methods differ by 47 per share on the same company.

8 Limits and Extensions

- **Terminal value dominates:** in the example it is three quarters of the value, so the result rests on WACC and g.
- **Flat assumptions:** one risk-free rate, one risk premium and one cost of debt for all companies. A low beta pushes the WACC toward g and inflates the terminal value.
- **Raw data:** Yahoo Finance fundamentals are not adjusted. There are no analyst estimates and no removal of one-off items.
- **Small peer group:** the median multiple comes from nine companies of different sectors, so it mixes unlike businesses.
- **Banks excluded:** the methods do not apply to JPM.

Three extensions would fix the largest limits. First, a sector-specific peer set with a size and growth adjustment, so that the multiple fits the firm. Second, a longer explicit forecast (10 years) with a margin path, which lets a high-growth company reach a steady state before the terminal value. Third, a reverse DCF that solves for the growth the price implies, since for the growth stocks the useful question is what the market assumes, not what the model returns.

9 Conclusion

Valuation outputs are conditional statements: given these cash flows and this discount rate, the value is this much. The grid for NVDA shows that for a growth stock the condition that matters is the growth path, which the model fixes in advance, so the most informative number would be the growth rate the price requires, which the toolkit does not yet compute.

References

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